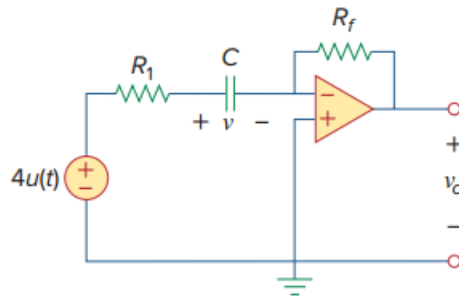


Problem

For the op amp circuit of Fig. 7.138, let $R_1 = 10\text{ k}\Omega$, $R_f = 30\text{ k}\Omega$, $C = 20\text{ }\mu\text{F}$, and $v(0) = 1\text{ V}$. Find v_o .

Fig. 7.138:



Step-by-step solution

Step 1 of 4

Refer to Figure 7.138 in the textbook.

Draw the circuit diagram for $t > 0$.

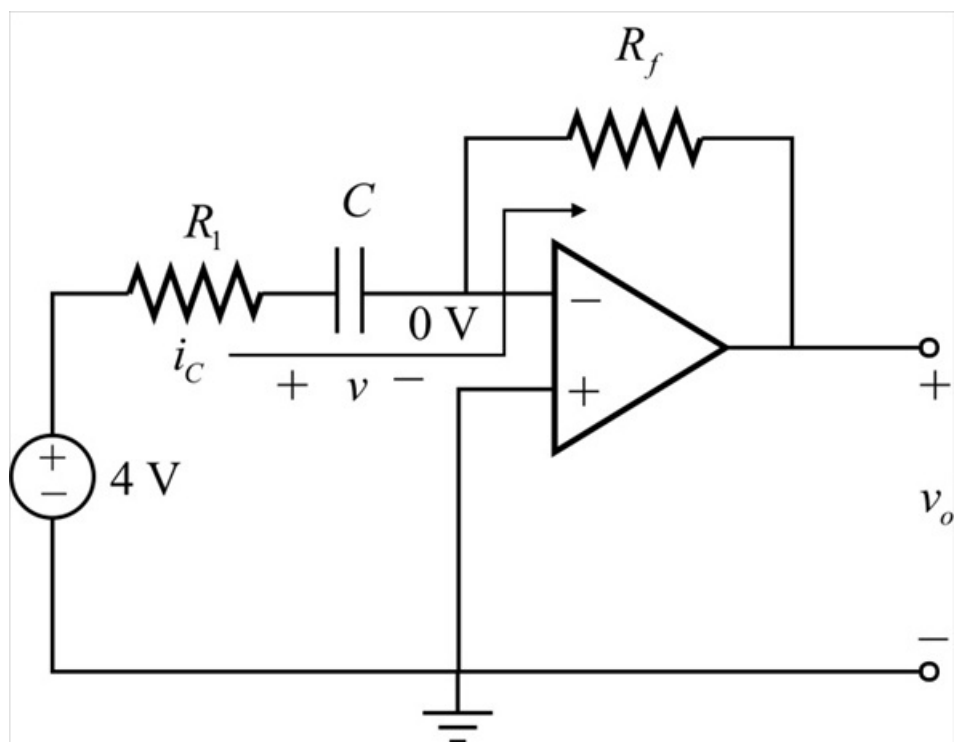


Figure 1

Step 2 of 4

As per virtual ground concept, the current through the terminals op-amp is zero. So, the current through the resistors R_1, R_f and capacitor is same.

The expression for the capacitor current is,

$$i_C = C \frac{dv}{dt}$$

At $t = \infty$, the capacitor acts as open circuit. So, the source voltage is dropped across the open circuited capacitor.

The final voltage of capacitor is,

$$v(\infty) = 4 \text{ V}$$

Calculate the time constant to calculate the capacitor voltage, τ .

$$\begin{aligned}\tau &= R_1 C \\ &= (10 \times 10^3)(20 \times 10^{-6}) \\ &= 0.2 \text{ sec}\end{aligned}$$

Step 3 of 4

Calculate the expression for capacitor voltage, $v(t)$.

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

Substitute 1 V for $v(0)$, 4 V for $v(\infty)$ and 0.2 sec for τ .

$$\begin{aligned}v(t) &= 4 + (1 - 4)e^{-\frac{t}{0.2}} \\ &= (4 - 3e^{-5t})u(t) \text{ V}\end{aligned}$$

Apply Kirchhoff's current law at the inverting terminal.

$$\frac{0 - v_o}{R_f} = i_C$$

$$C \frac{dv}{dt} = -\frac{v_o}{R_f}$$

$$v_o = -R_f C \frac{dv}{dt}$$

Step 4 of 4

Calculate the output voltage, $v_o(t)$

$$v_o(t) = -R_f C \frac{dv(t)}{dt}$$

Substitute $30 \text{ k}\Omega$ for R_f , $20 \text{ }\mu\text{F}$ for C and $(4 - 3e^{-5t})u(t)$ for $v(t)$.

$$\begin{aligned}v_o(t) &= -(30 \times 10^3)(20 \times 10^{-6}) \frac{d}{dt} [4 - 3e^{-5t}] \\ &= -(0.6) [0 - 3(-5)e^{-5t}] \\ &= -(0.6)(15e^{-5t}) \\ &= -9e^{-5t}u(t) \text{ V}\end{aligned}$$

Thus, the expression for the output voltage is $\boxed{-9e^{-5t}u(t) \text{ V}}$.